Analytical Cumulative Examination
December 5, 1998

Organized media; applications in capillary electrophoresis (CE).

1. Explain the origin of electroosmotic flow (EOF) in CE (hint: EOF increases with increasing pH).

2. Given below are data for EOF measured under various conditions for bare silica and an amine modified silica capillary. Explain these observations noting that + signifies flow toward the cathode.

	Electroosmotic velocity (mm/s) for $E = 4.0 \times 10^4$ V/m	
	pH 10	pH 2.5
Bare silica	+3.1	+0.2
Aminopropyl-modified silica	+1.8	−1.3

Data from K. Emoto, J. M. Harris, and M. Van Alstine, *Anal. Chem.* **1996**, *68*, 3751.

3. The electropherograms to the right show the separation of barbiturate compounds in the presence, and absence of α-cyclodextrin. The * identifies a chiral center. Explain the two electropherograms, and explicitly describe the origin for the separation observed when cyclodextrin is added to the running buffer. That is, describe the processes that resolve a single peak into a double peak.

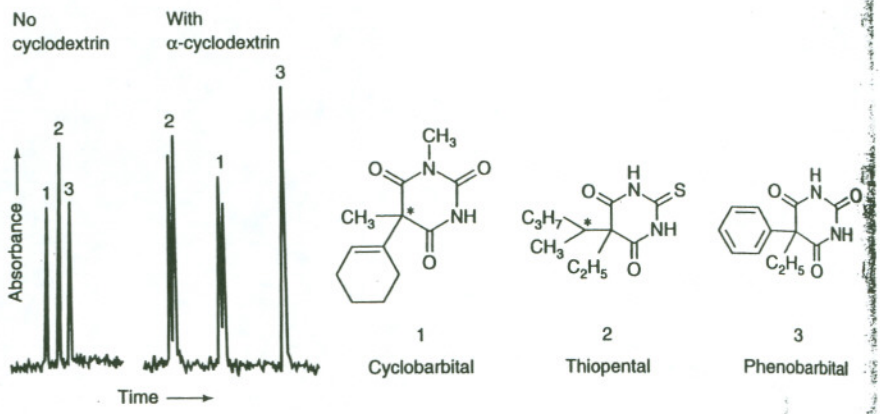
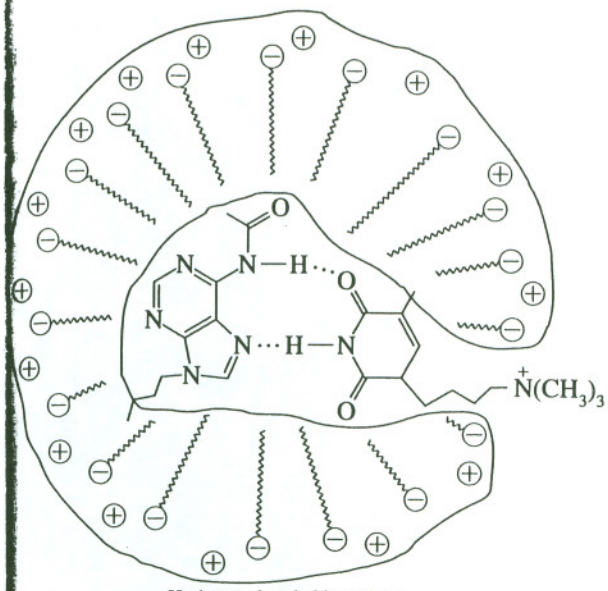


Figure 26-27 Separation of barbiturate enantiomers by micellar electrokinetic capillary chromatography. Separation in 20 mM borate buffer at pH 9.6 produces one peak for each compound. When 10 mM α-cyclodextrin (Box 24-1) is added, enantiomers of compounds 1 and 2 are resolved and all migration times change. Asterisks in the structures show chiral centers.

4. The water-soluble vitamins niacinamide (a neutral compound), riboflavin (neutral), niacin (an anion) and thiamine (a cation) were separated by micellar electrokinetic capillary chromatography in 15 mM borate buffer (pH 8.0) with 50 mM (above the CMC) sodium dodecyl sulfate (SDS). The migration times were: niacinamide (8.1 min.), riboflavin (13.0 min.), niacin (14.3 min.), and thiamine (21.9 min.). What would the order have been in the absence of SDS? Which compound is most soluble in the micelles?

5. It has been hypothesized that derivatives of the DNA bases thymine and adenine form a 1:1 hydrogen bonded complex inside an SDS micelle, as they do in DNA. To test this hypothesis, aliquots of 5.0 mM adenine derivative were mixed with aliquots of 5.0 mM thymine derivative in proportions given in the table below. The complex concentration was monitored via NMR. Are these results consistent with the hypothesis? Explain your answer. (hint: plot the data as [complex] versus mole fraction of one of the bases – interpret your plot to test the hypothesis).

Adenine volume (mL)	Thymine volume (mL)	Concentration of product (mM)
0.450	0.050	0.118 ± 0.009
0.400	0.100	0.202 ± 0.038
0.350	0.150	0.265 ± 0.021
0.300	0.200	0.307 ± 0.032
0.250	0.250	0.312 ± 0.060
0.200	0.300	0.296 ± 0.073
0.150	0.350	0.260 ± 0.122
0.100	0.400	0.187 ± 0.110
0.050	0.450	0.103 ± 0.104



Hydrogen-bonded base pair inside micelle with hydrocarbon tails anchoring the bases to the micelle